TABLE I
DESCRIPTION OF KEY NOTATIONS

Variable	Description
R, r	number, index of training rounds.
$\mathcal{N}, N, n$	set, number, index of clients.
M	number of sampled clients each round.
K	number of selected teachers each round.
$\mathcal{B}_n, \lvert\mathcal{B}_n\rvert$	dataset, dataset size of client n.
$\mathbf{D}_n$	local class distribution of client n.
$\mathbf{D}_{n,c}$	data proportion of the c-th class of client n.
$\Pi^{(r)}$	sampled client set in the r-th round.
$\pi_m^{(r)}$	index of the m-th sampled client in the r-th round.
$\mathcal{T}^{(r)}$	selected teacher client set for the r-th round.
$\mathcal{W}^{(r)}$	selected teacher model set for the r-th round.
$\mathbf{w}_m^{(r)}$	m-th trained local model in the r-th round.
$\pi_{t_k}^{(r-1)}$	k-th teacher client for the r-th round
$\mathbf{w}_{t_k}^{(r-1)}$	k-th teacher model for the r-th round
η	learning rate (or stepsize).
T_k, S	k-th teacher and student.
$\mathbf{D}^{T_k}, \mathbf{D}^S$	class distribution of k-th teacher client and student client.
g_k, h_k	NCKD's and TCKD's weight of teacher T_k.
$d(\cdot)$	distribution discrepancy metric.
z_c	output logits for class c.
p_c	predicted probability for class c.
τ	temperature.
$\mathbf{U}$	uniform distribution.
γ, β	trade-off coefficients of $\mathcal{L}_{NCKD}$ and $\mathcal{L}_{TCKD}$.

as teacher models to guide this round of training (Fig. 2 (II)), where K is the number of teachers.

During the r-th round, the sampled clients in the sequence $\Pi^{(r)}$ train the model one-by-one. Concretely, the m-th client $\pi_m^{(r)}$ receives the model trained by the previous client $\pi_{m-1}^{(r)}$. Then, it trains the model to learn from its local data while utilizing the teacher models in the subset $\mathcal{W}^{(r)}$ by performing the discrepancy-aware multi-teacher KD method based on the discrepancies between $\{\mathbf{D}_{\pi_{t_1}^{(r-1)}}, \cdots, \mathbf{D}_{\pi_{t_k}^{(r-1)}}, \cdots, \mathbf{D}_{\pi_{t_K}^{(r-1)}}\}$ and $\mathbf{D}_{\pi_m^{(r)}}$ (Fig. 2 (III)), where $\mathbf{D}_{\pi_{t_k}^{(r-1)}}$ is the k-th teacher client's local class distribution. Here, the server obtains the local class distributions from clients and computes discrepancies. As we primarily focus on addressing catastrophic forgetting in SFL, we assume the server is trustworthy. We can address potential privacy issues using similar methods to [55]. The training performs Stochastic Gradient Descent as follows:

$$\mathbf{w}_m^{(r)} = \mathbf{w}_{m-1}^{(r)} - \eta \nabla \mathcal{L}(\mathbf{w}_{m-1}^{(r)}, \boldsymbol{\xi}_{\pi_m^{(r)}}), \tag{2}$$

where $\boldsymbol{\xi}_{\pi_m^{(r)}}$ is the mini-batch sampled from $\mathcal{B}_{\pi_m^{(r)}}$, η is the learning rate. After training on the client $\pi_m^{(r)}$ is completed, the model $\mathbf{w}_m^{(r)}$ is passed to the server and the next client $\pi_{m+1}^{(r)}$ in the sequence.

IV. Discrepancy-Aware Multi-Teacher KD

In this subsection, we propose the discrepancy-aware multi-teacher KD method for the SFedKD framework. Note that multiple teacher models will be selected to guide the current round of training, and these teacher models contribute to each client's training differently. Although traditional multi-teacher KD methods [46], [48] can characterize personalized teacher contributions, they fail to effectively balance training effectiveness and catastrophic forgetting mitigation due to their rough weight strategies. Therefore, we extend the single-teacher Decoupled Knowledge Distillation [29] approach to our multi-teacher setting. Moreover, we utilize its two decoupled components, Non-target Class Knowledge Distillation (NCKD) and Target Class Knowledge Distillation (TCKD), and assign distinct weights to them respectively based on the class distributional discrepancies between the teacher clients' and the student client's local data. Through this fine-grained weighting strategy, the former component specifically addresses catastrophic forgetting mitigation, while the latter enhances model training efficacy, collectively establishing an optimal balance to improve overall training performance.

First, we extend the components NCKD and TCKD to our multi-teacher setting. We leverage the Kullback-Leibler (KL) Loss [56], which mathematically equals the cross-entropy between the teacher and student distributions minus the intrinsic entropy of the teacher distribution. This formulation enables the student to better capture the uncertainty and structural knowledge embedded in the teachers. Specifically, NCKD represents the weighted KL-Divergence between the teacher and student models' softmax probability vectors over all non-target classes, formulated as:

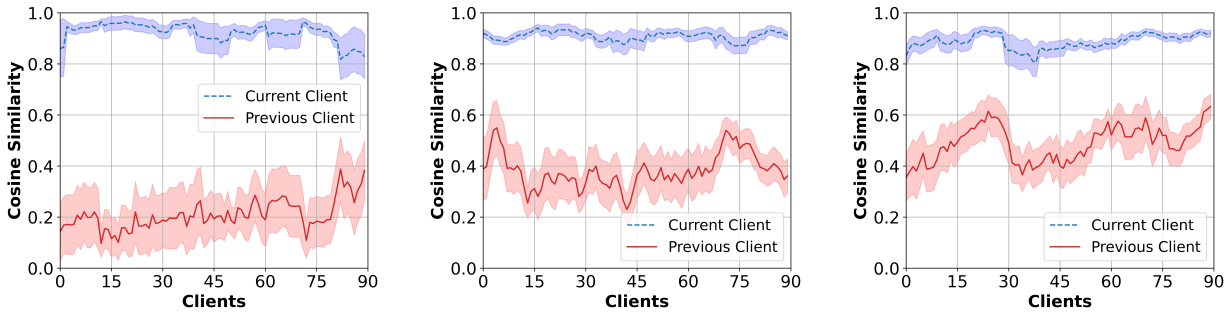
$$\mathcal{L}_{NCKD} = \sum_{k=1}^{K} g_k \sum_{c=1,\underline{\mathbf{c}\neq\mathbf{t}}}^{C} \tilde{p}_c^{T_k} \log\left[\frac{\tilde{p}_c^{T_k}}{\tilde{p}_c^{S}}\right], \text{ where}$$
$$\tilde{p}_c^{T_k} = \frac{\exp(z_c^{T_k}/\tau)}{\sum_{\tilde{\mathbf{c}}\neq\mathbf{t}}^{C}\exp(z_{\tilde{c}}^{T_k}/\tau)}, \ \tilde{p}_c^{S} = \frac{\exp(z_c^{S}/\tau)}{\sum_{\tilde{\mathbf{c}}\neq\mathbf{t}}^{C}\exp(z_{\tilde{c}}^{S}/\tau)}, \forall c \neq t, \tag{3}$$

where $\mathbf{z} = [z_1, \cdots, z_t, \cdots, z_C]$ is the logits output of training sample, t is the target class, τ is temperature, T_k is the k-th teacher, g_k represents the corresponding weight, K is the number of teachers, and S denotes the student. TCKD is the weighted KL-Divergence between the binary probabilities of the target class and all the other non-target classes from teacher and student models, formulated as:

$$\mathcal{L}_{TCKD} = \sum_{k=1}^{K} h_k(p_t^{T_k}\log\left[\frac{p_t^{T_k}}{p_t^{S}}\right] + p_{\tilde{t}}^{T_k}\log\left[\frac{p_{\tilde{t}}^{T_k}}{p_{\tilde{t}}^{S}}\right]), \text{ where}$$
$$p_t^{T_k} = \frac{\exp(z_t^{T_k}/\tau)}{\sum_{\mathbf{c}=\mathbf{1}}^{C}\exp(z_c^{T_k}/\tau)}, \ p_t^{S} = \frac{\exp(z_t^{S}/\tau)}{\sum_{\mathbf{c}=\mathbf{1}}^{C}\exp(z_c^{S}/\tau)}$$
$$p_{\tilde{t}}^{T_k} = \frac{\sum_{\tilde{\mathbf{c}}\neq\mathbf{t}}^{C}\exp(z_{\tilde{c}}^{T_k}/\tau)}{\sum_{\mathbf{c}=\mathbf{1}}^{C}\exp(z_c^{T_k}/\tau)}, \ p_{\tilde{t}}^{S} = \frac{\sum_{\tilde{\mathbf{c}}\neq\mathbf{t}}^{C}\exp(z_{\tilde{c}}^{S}/\tau)}{\sum_{\mathbf{c}=\mathbf{1}}^{C}\exp(z_c^{S}/\tau)}. \tag{4}$$

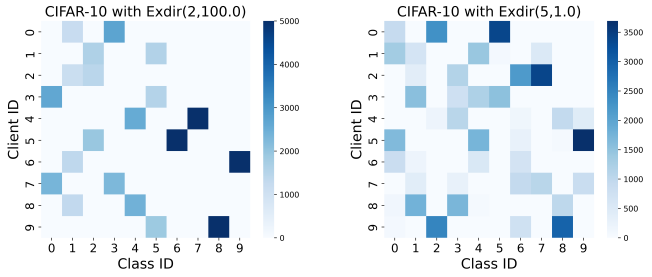
Here, h_k is the TCKD's weight of teacher T_k.

Second, we design personalized weight values to g_k in Eq. (3) and h_k in Eq. (4) based on the discrepancy between the local class distribution of each teacher client and the student client. For conciseness, we use $\mathbf{D}^{T_k}$ and $\mathbf{D}^S$ to represent the local class distribution of the k-th teacher client and the student client where the model is performing distillation, respectively.



(a) dir(0.1) (b) dir(0.5) (c) dir(1.0)

Fig. 3. Cosine similarity between the class-wise test accuracy vector and the class distributions of the current and previous clients.



(a) Exdir(2,100.0) (b) Exdir(5,1.0)

Fig. 4. Two data distributions across clients visualized on CIFAR-10, where the color depth of each square represents the number of training samples per class for that client.

more similar. Note that Eq. (6) minimizes the distance between the aggregated local class distributions of the K teacher clients and $\mathbf{U}$, where $\mathbf{U}$ denotes the uniform distribution. Actually, since $\mathbf{U}$ represents the optimal state of maximizing class coverage while maintaining balance across classes, our distance minimization objective is effectively equivalent to a maximum coverage problem with a balance constraint. Therefore, this problem constitutes a variant of the maximum coverage problem, which is NP-hard. There is no polynomial-time optimal algorithm for this problem, so we develop a greedy teacher selection algorithm in Algorithm 1 to solve it. We select the optimal client π_{t_k} that minimizes the discrepancy between the aggregated class distribution $\mathbf{D}_{agg}$ and uniform distribution $\mathbf{U}$ in the k-th iteration until reaching the target teacher number K (Lines 2-5).

Through the above mechanism, the selected teacher clients exhibit a complementary relationship, as their aggregated class distribution approximates a uniform distribution, ensuring comprehensive coverage of the class knowledge space, which thereby enhances the generalization capability of the student model and the convergence of SFL. In addition, our selection strategy actively prunes redundant teachers, thereby preventing knowledge dilution while reducing communication and computation costs.

VI. Overall Loss Function and Algorithm

In summary, our loss function can be summarized as:

$$\mathcal{L} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{NCKD} + \beta \mathcal{L}_{TCKD}, \tag{7}$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss function that measures the difference between the prediction and the ground truth labels, γ and β denote the corresponding trade-off coefficients, $\mathcal{L}_{NCKD}$ and $\mathcal{L}_{TCKD}$ are respectively defined in Eq. (3) and Eq. (4).

For a better understanding of our method, we show the pseudocode of our framework SFedKD in Algorithm 2. At the beginning of each round, the server samples a client sequence and executes the teacher selection mechanism to get teachers (Lines 3-6). Then, the sampled clients train the model according to the overall loss function defined in Eq. (7) in sequence (Lines 7-9). At last, the final client outputs the ultimate global model (Lines 10-11).

VII. Experiments

A. Experimental Setup

1) Dataset: We conduct thorough experiments on five public benchmark datasets, including Fashion-MNIST [57], CIFAR-10 [58], CINIC-10 [59], CIFAR-100 [58], and HAM10000 [60]. Fashion-MNIST comprises of 28 × 28 grayscale images of 70,000 fashion products. The CIFAR-10 dataset consists of 50,000 training images and 10,000 testing images, each with a size of 3 × 32 × 32. CINIC-10 is an extension of CIFAR-10 via the addition of downsampled ImageNet [61] images. CIFAR-100 has the same format as CIFAR-10 with 100 classes of images. The HAM10000 dataset is an image dataset used for skin lesion classification in the medical field with 7 classes.

2) Model Architecture: We use a LeNet-5 [62] for Fashion-MNIST, an AlexNet [63] for CIFAR-10 and CINIC-10, and a CNN for other datasets. For Fashion-MNIST, we use a LeNet-5 with two 5 × 5 convolutional layers, each followed by ReLU activation and max pooling, and three fully connected layers. We use an 8-layer AlexNet with a size of 136MB for CIFAR-10 and CINIC-10, which includes five 3 × 3 convolutional layers with ReLU and max pooling, followed by three fully connected layers with dropout. We train models on CIFAR-100 using a 5-layer CNN with two 5 × 5 convolutional layers, each followed by ReLU activation and max pooling, and three fully connected layers. The model for HAM10000 is a CNN consisting of three 3 × 3 convolutional layers with ReLU activation and max pooling, and two fully connected layers with dropout regularization.